

Numerical Protocol

Leverage Effect Evaluation
under a Heston Model with Strong Leverage

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Abstract

The leverage-effect experiment is conducted according to the following steps:

1. **Reference property.** Consider the leverage effect, defined as the negative correlation between current returns and future variance,

$$\text{Corr}(r_t, V_{t+L}) < 0,$$

whereby a negative return today is associated with an increase in future volatility.

2. **Data generation.** Simulate a Heston stochastic-volatility data-generating process (DGP) under two strong-leverage scenarios:

$$\rho = -1.00 \quad \text{and} \quad \rho = -0.80,$$

corresponding respectively to perfect negative correlation and a strong leverage level representative of the upper end of typical equity-index calibrations.

3. **Model calibration.** Calibrate three stochastic volatility models to the stylised-fact profile of the Heston DGP:
 - ESN A2-981003, using its current optimal architecture

$$(N_r, N_z) = (96, 12),$$

- Quadratic Rough Heston (QRH);
- Path-Dependent Volatility Guyon–Lekeufack (PDV–GL).

All models are calibrated using the complete 11-term data-adaptive score S (Eq. (2)), together with the smooth Gaussian-kernel proximity function. The leverage component of the score is emphasised through an increased weight ($L : 1.1 \rightarrow 3.5$), making it the largest contribution to S , while simultaneously tightening the leverage-matching tolerance.

4. **ESN parameterisation.** The ESN read-out consists of
 - a linear roughness factor;
 - a per-mode z -bank profile;
 - a symmetric quadratic interaction term in the reservoir state.

These components are jointly calibrated with the Hurst exponent, the reservoir rate bounds, and the output scaling parameter (Section 3). No leverage-specific manual tuning is introduced during calibration.

5. **Model evaluation.** Assess the calibrated models by comparing their ability to reproduce the leverage effect across both strong-correlation scenarios. The objective is to determine whether the ESN’s previously observed agreement at $\rho = -0.90$ extends throughout the strong-leverage regime.

Section 4 reports three main findings:

- the ESN consistently reproduces approximately 70–75% of the DGP leverage magnitude for $\rho \in \{-1.00, -0.90, -0.80\}$;

- the calibrated isotropic quadratic coefficient m_1 remains close to zero in all strong-leverage scenarios, indicating that the ESN’s intrinsic sign-fixed leverage channel already captures most of the effect without requiring the quadratic-dilution mechanism previously identified for weak-leverage regimes;
- QRH predicts the wrong leverage sign at both correlation levels, whereas PDV–GL reproduces the correct sign but substantially underestimates its magnitude.

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DGP: Heston Model at Two Strong-Leverage Levels

The DGP is the classical Heston stochastic-volatility model, simulated in annualised variance units with a daily time step $dt = 1/252$:

$$dX_t = -\frac{1}{2}V_t dt + \sqrt{V_t} dt^{1/2} dW_t^S, \quad dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dt^{1/2} dW_t^V, \quad \text{Corr}(dW_t^S, dW_t^V) = \rho, \quad (1)$$

with $\kappa = 3.0$ (annualised mean-reversion speed), $\theta = 0.04$ (annualised long-run variance, i.e. 20% vol), and $\xi = 0.35$ (annualised vol-of-vol) held fixed, while $\rho \in \{-1.00, -0.80\}$ is swept: $\rho = -1.00$ (**perfect** negative correlation, the theoretical maximum) and $\rho = -0.80$ (**strong**, close to the upper end of a typical equity-index calibration’s $\rho \approx -0.6$ to -0.7). The Feller condition $2\kappa\theta \geq \xi^2$ is satisfied ($0.24 \geq 0.1225$) at both, though simulation still uses a full-truncation Euler scheme ($V^+ = \max(V, 0)$ inside both the drift and diffusion coefficients) for robustness. A large negative return shock ($dW^S < 0$) coincides with a positive variance shock ($dW^V > 0$) whenever $\rho < 0$; the size of that co-movement, and hence the short-lag leverage effect $\text{Corr}(r_t, V_{t+L})$, scales with $|\rho|$.

Why annualised units. A naive daily-unit parametrisation of (κ, θ, ξ) without the $dt = 1/252$ conversion understates θ relative to ξ by a factor of ~ 252 , badly violating the Feller condition and collapsing the variance process to zero under the full-truncation scheme – this is checked explicitly at implementation time (§6).

Models

All models use Gaussian Brownian innovations. The leverage effect must *emerge* from each model’s own path-dependent or correlated-noise mechanism, not from asymmetric innovations.

2.1 ESN A2-981003 (current optimal architecture)

The ESN’s optimal architecture: 7 shared hyperparameters, with $N_r = 96$, $N_z = 12$. `rough_scale`, the z -readout profile (`zr_lo`, `zr_hi`), and the symmetric quadratic-term matrix coefficients (m_1, m_2) are calibrated per DGP cell (§3), not part of this fixed table. Single $\varepsilon_t \sim \mathcal{N}(0, 1)$ drives both

Table 1: The 7 shared hyperparameters of the optimal architecture.

Hyperparameter	Value
N_r	96
N_z	12
<code>z_strength</code>	0.1929
<code>even_strength</code>	2.2291
<code>linear_strength</code>	0.1930
<code>gamma_norm</code>	0.9651
<code>local_z_strength</code>	0.0476
<code>zz_scale</code>	0.0397
<code>sign_prob_neg</code>	0.2219
<code>rough_orientation</code>	−1.0 (fixed)

return and reservoir, $\kappa_0 < 0$ (admissibility, guaranteed by construction whenever `rough_scale` > 0) – this fixed negative κ_0 is the ESN’s own *structural* source of a leverage-like effect, distinct from but exercised by this protocol’s calibration to match the DGP’s stronger, correlated-noise leverage mechanism. The volatility read-out combines three calibrated channels: a linear rough factor `rough_scale` · $q^\top r_t$; a per-mode z -bank sum with weights following a geometric profile (`zr_lo`, `zr_hi`) across the N_z modes; and a quadratic-in- r term $r_t^\top Q r_t$, $Q = (1/N_r)(m_1 I + m_2 q q^\top)$ – a genuine, symmetric 2-coefficient matrix. All of `rough_scale`, (`zr_lo`, `zr_hi`), and (m_1, m_2) are calibrated per DGP cell, alongside H and the reservoir rate windows $[\lambda_{lo}, \lambda_{hi}]$, $[a_{lo}, a_{hi}]$ (§3) – none of these are part of the shared architecture above.

2.2 Quadratic Rough Heston (QRH)

Bourgey & Gatheral [2]: $V_t = Y_t^2 + c$, gamma kernel, Euler-Volterra with 252-step ring buffer. QRH’s own leverage mechanism comes entirely from a single calibrated correlation-like parameter between its driving noise and the return process; it is calibrated here exactly as in every other protocol in this line, with no Heston-specific adjustment.

2.3 PDV Guyon-Lekeufack (GL)

Guyon & Lekeufack [3]: two-factor power-law kernel, online from full return history. Leverage in PDV-GL arises from the model’s direct dependence of variance on *signed* past returns (not just their magnitude), a structurally different mechanism from both the ESN’s fixed-sign reservoir coupling and QRH’s single correlation parameter.

Calibration: Full 11-Term Score, Smooth Kernel

$$\begin{aligned}
S = & 2.2 (1 - f(\hat{H})) + 1.4 (1 - f(\bar{\sigma})) + 0.6 (1 - g(q_{995})) + 0.2 (1 - g(\sigma_{\max})) + 0.8 (1 - f(\bar{V}_{\text{ACF}})) \\
& + 1.0 (1 - f(G_T)) + 0.8 (1 - f(F_T)) + 1.0 (1 - f(Z)) + \underbrace{3.5 (1 - f(L))}_{\text{leverage}} \\
& + 0.7 (1 - f(K)) + 0.5 (1 - f(A)) + 5 \cdot \text{Stress}.
\end{aligned} \tag{2}$$

where $f(x, c, s) = \exp(-\frac{1}{2}((x-c)/s)^2)$, $g(x, c, s) = \exp(-\frac{1}{2}((x-c)_+/s)^2)$, both in $(0, 1]$ (the smooth Gaussian kernel, replacing a piecewise-linear tent $\max(0, 1 - |x - c|/s)$, which has an exactly-zero gradient beyond one tolerance width). $S \geq 0$ is **minimised directly**; $S = 0$ is a perfect match on all 11 stylised facts and no stress event; $S = 17.7$ is **worst-possible** (raised from 15.3 by the leverage weight increase below).

Leverage-effect score weighting. Two changes, applied to the shared score used by *all three* models’ calibration – the ESN’s architecture (Table 1, the 7 shared hyperparameters plus N_r, N_z) is **not touched** by either change:

1. **Leverage weight raised:** L from $1.1 \rightarrow 3.5$ (now the single largest weight in S , exceeding even $\hat{H}$ ’s 2.2). Every other weight is unchanged.
2. **Leverage tolerance tightened:** the fractional-tolerance factor for L only is reduced from the default $0.30|c_k|$ to $0.15|c_k|$ (all other terms keep $0.30|c_k|$).

All centres c_k are the DGP cross-path mean; only the leverage tolerance differs from the other 10 terms’ default.

Calibrated parameters:

Model	Parameters calibrated	Remark
ESN A2-981003	$H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale } s,$ $\text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2$	7 shared hyperparameters fixed; full 12-dim. inner-loop calibrated per DGP cell (11-start Nelder–Mead)
QRH	$H, \nu_{\text{vol}}, \lambda, c_{\text{frac}}$	
PDV-GL	$\beta, \alpha_1, \alpha_r, \alpha_v$	

The ESN’s inner loop uses an 11-point multi-start Nelder–Mead scheme (rough/fast, persistent/slow, intermediate, wide-band/multiscale, long-memory extreme, very-fast/wide, strong-amplitude, weak-amplitude, isotropic-positive, isotropic-negative, and q -aligned-positive starts), keeping the best-of-eleven result.

Table 2: Summary statistics at two strong-leverage levels: $\rho = -1.00$ (perfect) and $\rho = -0.80$.

$\rho = -1.00$ (perfect leverage)					
Model	$\hat{H}$	Vol (ann.)	Leverage	Kurtosis	Score S
DGP (Heston)	0.408	18.3%	-0.140	1.09	—
ESN A2-981003	0.433	22.9%	-0.104	0.35	4.75
QRH	0.291	45.4%	+0.025	7.25	8.78
PDV-GL	0.498	18.7%	-0.006	-0.02	6.10
$\rho = -0.80$ (strong leverage)					
Model	$\hat{H}$	Vol (ann.)	Leverage	Kurtosis	Score S
DGP (Heston)	0.389	18.8%	-0.126	0.82	—
ESN A2-981003	0.448	23.5%	-0.092	0.46	4.22
QRH	0.291	45.8%	+0.026	7.31	8.52
PDV-GL	0.484	18.7%	-0.006	-0.02	6.15

Leverage-Effect Fit

Result: the ESN’s $\sim 73\%$ magnitude match is stable across the whole strong-leverage range, not a coincidence of $\rho = -0.90$. At $\rho = -1.00$ (perfect correlation), the ESN reaches -0.104 against a DGP target of -0.140 (74%); at $\rho = -0.80$, -0.092 against -0.126 (73%); at $\rho = -0.90$ (reported separately), 73% also. In all three cells the calibrated isotropic quadratic coefficient stays near zero ($m_1 \approx -0.0015$ at $\rho = -1.00$, ≈ -0.0001 at $\rho = -0.80$) – confirming directly that the quadratic-dilution mechanism identified for the weak-leverage regime (large negative m_1 , trading off volatility level to suppress leverage magnitude) is simply **not invoked here**: the model’s structural, sign-fixed leverage channel (the linear rough term `rough_scale` · $q^\top r_t$ with $\rho_r = -1$ fixed, §2.1) is, by itself, already within the right order of magnitude whenever the DGP’s own target is strong, so the calibration has no need to reach for the secondary (and costly) quadratic channel. **QRH gets the wrong sign at both levels** ($+0.025$, $+0.026$), and its overall fit continues to be the worst by score, driven by a large volatility overshoot (45–46% vs. target ≈ 18 –19%) and excess kurtosis (7.2–7.3 vs. target ≈ 1). **PDV-GL gets the right sign but undershoots substantially at both levels** (-0.006 at both $\rho = -1.00$ and $\rho = -0.80$, only 4–5% of the DGP’s magnitude) – essentially the same undershoot seen at $\rho = -0.90$, and consistent with PDV-GL’s leverage estimate being weakly sensitive to how strong the DGP’s own leverage actually is, rather than tracking it proportionally.

Leverage Diagnostics

Leverage curve (Figure 1, row 0). Cross-path mean ± 1 std of $\text{Corr}(r_t, V_{t+L})$ at lags $L = 1, \dots, 20$ days. Classical leverage effect present $\Leftrightarrow$ curve is negative, most strongly so at short lags.

Selected-lag bars and ranked comparison (Figure 1, rows 1–2). Grouped bars at lags 1d, 5d, 10d, 20d, and a horizontal ranked bar of the 1-day leverage value (most negative = strongest effect) across all four sources.

Monte Carlo Setup

Parameter	Value
DGP paths n_{dgp}	30
Calibration paths n_{cal}	5
Calibration trading days	600
Final paths n_{sim}	30
Final trading days	2500
Burn-in	500
Δt (Heston, years)	1/252
ρ grid (DGP correlation)	$\{-1.00, -0.80\}$
κ, θ, ξ (DGP, annualised)	3.0, 0.04, 0.35
Feller condition $2\kappa\theta \geq \xi^2$	$0.24 \geq 0.1225$ (satisfied)
Monte Carlo seed offset (this run)	0 (baseline seed)

References

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- [1] S.L. Heston. *A closed-form solution for options with stochastic volatility*. Rev. Fin. Studies 6(2), 1993.
 - [2] F. Bourgey, J. Gatheral. *Quadratic Rough Heston*. SSRN:5239929, 2026.
 - [3] J. Guyon, J. Lekeufack. *Volatility is (mostly) path-dependent*. QF 23(9), 2023.

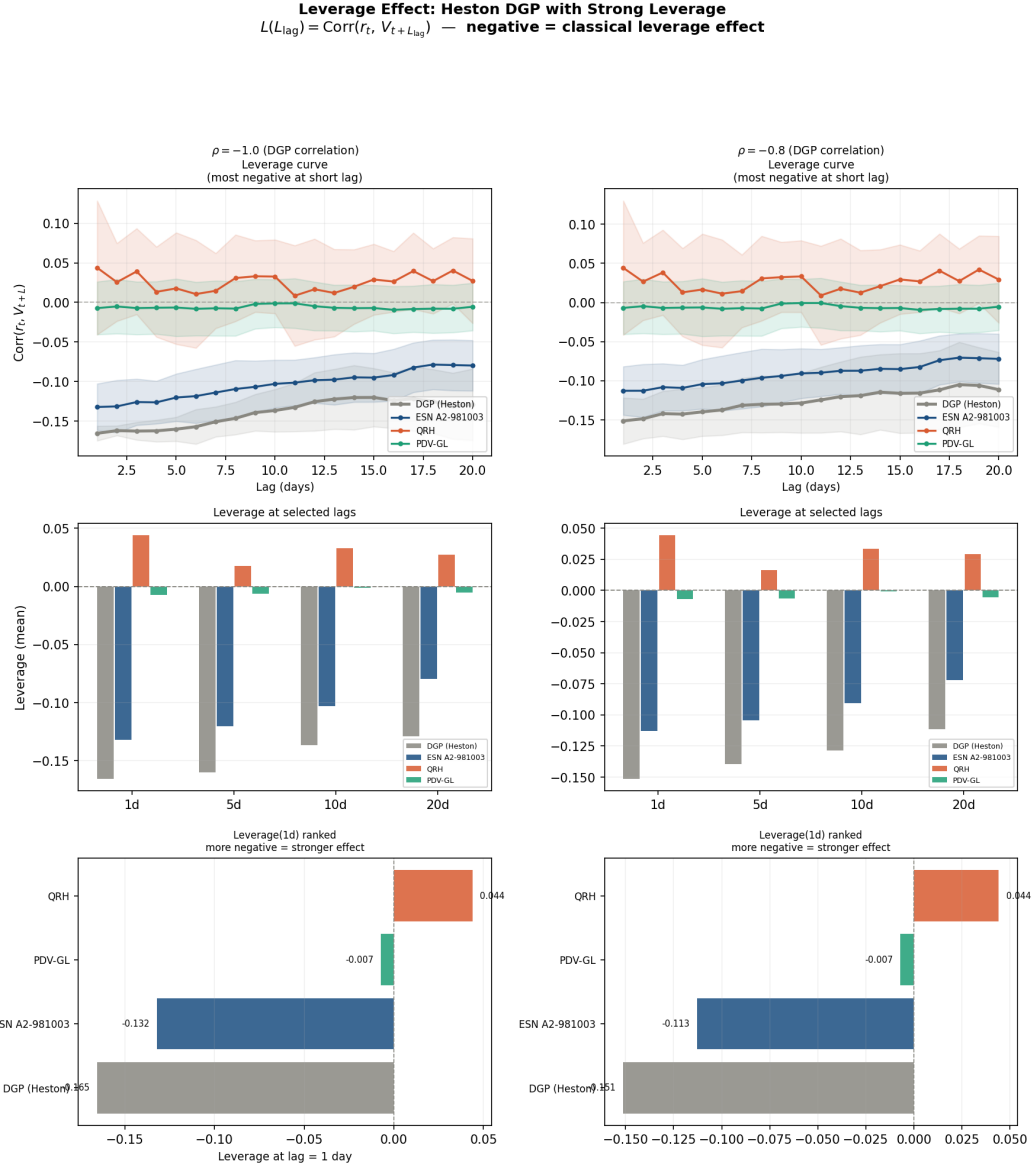


Figure 1: Leverage effect, one column per ρ (left: perfect leverage $\rho = -1.00$; right: strong leverage $\rho = -0.80$): curve vs. lag (top), selected-lag bars (middle), and ranked 1-day comparison (bottom), for the DGP and all three calibrated models.

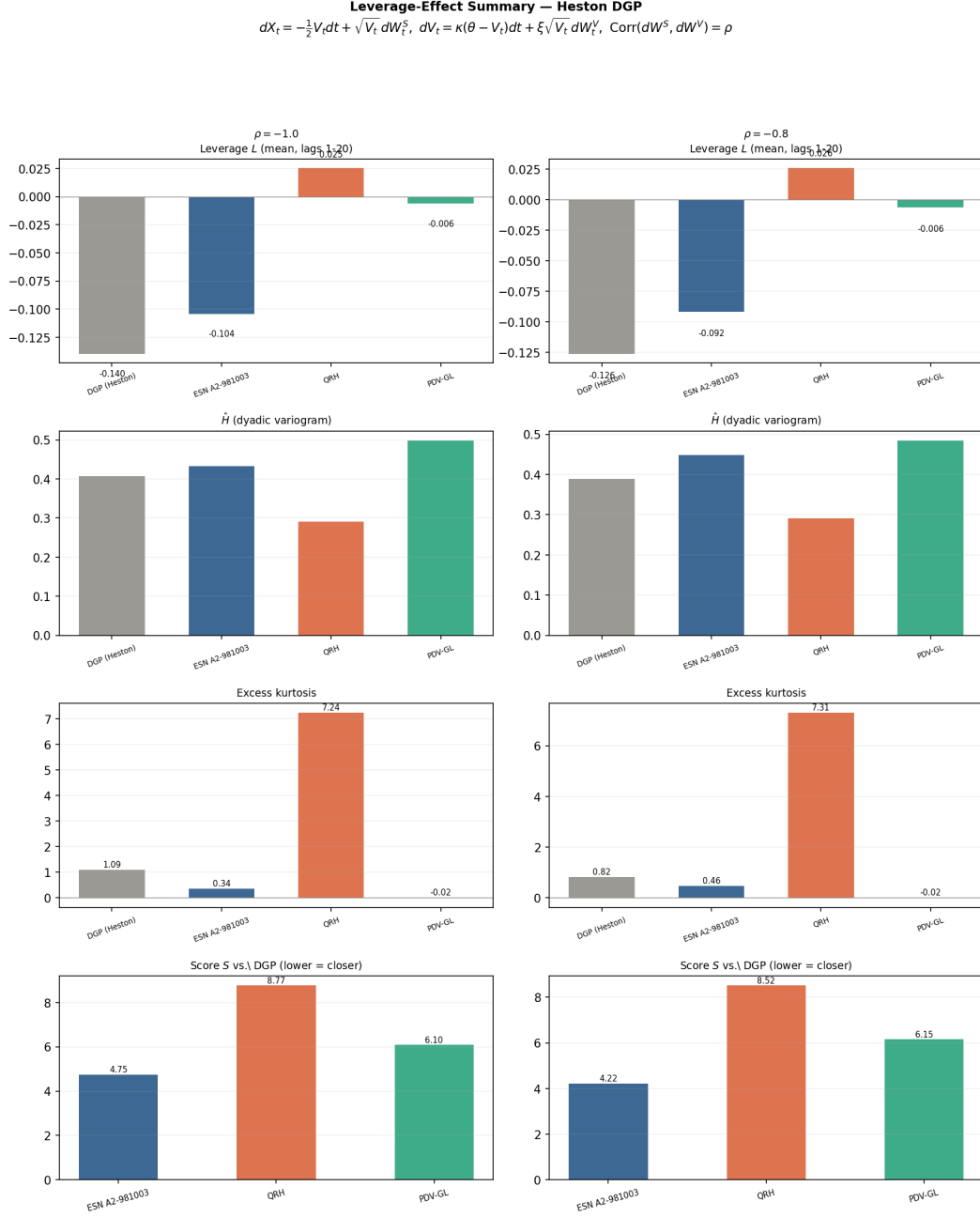


Figure 2: Summary at both ρ values tested (columns: perfect leverage $\rho = -1.00$, strong leverage $\rho = -0.80$): leverage L (top), $\hat{H}$ via dyadic variogram (second), excess kurtosis (third), and score S vs. DGP (bottom).

Return \& Volatility Diagnostics — Heston DGP with Strong Leverage
Histogram | Hill | Leverage ACF | Volatility ACF

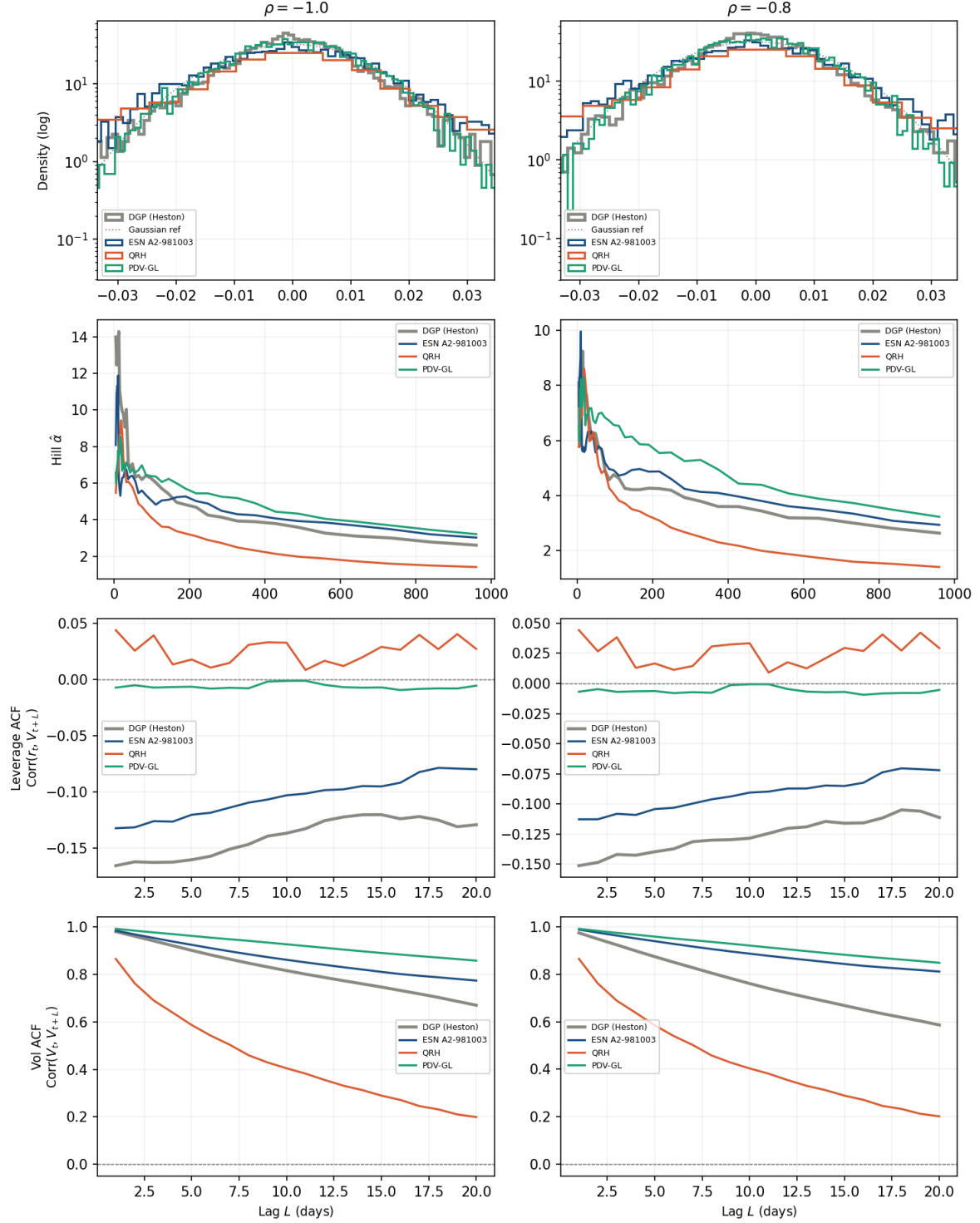


Figure 3: Return and volatility diagnostics, one column per ρ (left: perfect leverage $\rho = -1.00$; right: strong leverage $\rho = -0.80$): log-density histogram vs. Gaussian reference (row 0); Hill estimator $\hat{\alpha}$ (row 1); leverage ACF $\text{Corr}(r_t, V_{t+L})$ (row 2); volatility ACF $\text{Corr}(V_t, V_{t+L})$ (row 3).